

Working with Codex

In crew 1.0, the interactive Claude session implements and Codex reviews. Both agents read the same instructions, and the same context hook can run under both. This guide explains how to set up the Codex half and what is proven about it so far.

What Each Side Reads

Surface	Claude Code	Codex
AGENTS.md (80 lines or fewer, generated)	through @AGENTS.md in CLAUDE.md	natively
.claude/rules/<subsystem>.md (paths:-scoped)	yes, when a matching file is read	no (Codex has no path-scoped rules)
crew's context hook	plugins hooks.json	.codex/hooks.json (generated)
CLAUDE.md	natively	fallback via project_doc_fallback_filenames

Codex has no path-scoped rules, so its hook's PostToolUse slice is the only per-file channel.

Generate the Files

Run these from the repository root. Each command writes nothing if the files are already current. With --check, it writes nothing at all and exits 1 on drift.

```
S="<crew plugin root>/hooks/scripts"

python3 "$S/crew_instructions.py" agents      # AGENTS.md; only the crew:keep block
is yours

python3 "$S/crew_instructions.py" rules      # .claude/rules/<subsystem>.md
from .crew/codemap/

python3 "$S/crew_instructions.py" codex      # .codex/hooks.json and
.codex/config.toml
```

```
python3 "$S/crew_instructions.py" codex --check
```

- `.codex/hooks.json` is built from the same hook table as crew's Claude registration (`H00K_TABLE` in `crew_instructions.py`). Each entry has a `command` (bash) and a `commandWindows` (PowerShell, `-Harness codex`). The paths inside are absolute and belong to one machine, so regenerate the file on each machine rather than committing it.
- `.codex/config.toml` sets `project_doc_fallback_filenames = ["CLAUDE.md"]` only. **No [profiles.*] table** -- MEASURED on codex-cli 0.154.0: `profiles` is on Codex's own project-local config denylist and is stripped from this file every time it loads, trusted or not, so a review/work profile defined here would never apply. Codex reads project config only for a **trusted** project, and only that project-config subset survives the strip.

Review and Work

There are no project-scoped review/work profiles to select with `--profile` -- pass the settings explicitly instead:

```
codex exec --json --sandbox read-only "<review prompt>" # the reviewer
(review_run.py's command_for)

codex exec --sandbox workspace-write "<task>" # only if you choose
Codex to implement
```

If you want named review/work profiles for convenience, they are a **user-level** concept, not a project one: `codex exec --help` (codex-cli 0.155.1) documents `--profile NAME` as layering `$CODEX_HOME/NAME.config.toml` on top of the base user config -- a file under `$CODEX_HOME`, never this project's `.codex/config.toml`. Create `$CODEX_HOME/review.config.toml` / `$CODEX_HOME/work.config.toml` yourself and pass `--profile review/--profile work` against those.

A review that exits non-zero, prints nothing, times out or ends in a failed turn is **INCOMPLETE**, never **CLEAN**. That is a tool failure, and the round is refunded automatically, up to two per plan (for example when Codex answers with its usage-limit error event). What crew relies on from Codex's `--json` stream is in `plugin/crew/docs/external-tool-formats.md`.

The review prompt Codex reads ends with a **development standards checklist**: the rules and self-check questions of every standard that applies to the change (crew's generic set, any

matching per-language set, and the repository's `.crew/standards.md` overlay). The author's own self-check answers are withheld from the prompt, so Codex judges each standard's applicability itself, and the prompt says the list does not bound the review: a defect outside it is reported the same way. Withheld from the prompt is not hidden: the answers stay in

`.work/tickets/<id>/selfcheck.md`, which Codex, with read access to the checkout, could open; the prompt never names that file. For a ticket with an approval receipt (or one whose receipt cannot be proven absent), `review_run.py` will not launch Codex, or reserve a round, until the author's self-check is complete and stamped for the exact bundle Codex is about to read. A ticket with no approval receipt is told the gate does not apply, and during a declared incident (`/crew:emergency`) the gate stands down and logs the skip. The self-check is asked for last: a CLEAN receipt that already covers the bundle answers CLEAN without it, and a tree the verify gate has not passed is refused (exit 5) before it.

When parallel lanes share one clone and its **merge train** is armed (`crew_train.py arm`, L-0520), `review_run.py` also takes the train before it launches Codex or reserves a round: after the CLEAN-receipt short-circuit and the verify gate, before the self-check. A round for a ticket whose Touch overlaps a ticket already holding the train (or queued ahead of it on the same base), or whose base moved in its Touch paths and was not merged in, exits **6** with no round spent and names the blocker and the colliding paths; `crew_train.py status` shows the queue. Codex never sees a tree the train would not land. A clone that was never armed reviews exactly as before. When a catch-up merge (`crew_train.py catch-up`) replayed a recorded conflict resolution through git rerere, the prompt Codex reads lists each replayed file under `== Catch-up merges (rerere) ==` as a change to review.

When Codex Hits a Usage Limit

Having codex on PATH does not mean it can review: a logged-out, rate-limited or out-of-credits Codex is installed and fails at the first call. So `/crew:review` makes one minimal real call first, before it reserves a round, with the round's own model and effort:

```
python3 "$S/review_run.py" --root . --ticket <id> --scratch <dir> --provider codex \
--model <model> --effort <effort> --probe
```

It prints `PROBE=<outcome>` and `PROBE_DETAIL=<the answer or the quoted error>` and exits 0 (ok), 5 (limited), 6 (failed) or 7 (unknown: no answer within 120 s). Nothing is reserved either way. limited means the call failed and its error matched one of Codex's own limit messages (usage

limit, out of credits, spend cap, rate limit, quota, a plan without Codex, or a retry limit on HTTP 429), listed with their `error.rs` lines in `hooks/scripts/review_limit.py`. On limited the round runs on the Claude reviewer - even when `qa.provider` pins `codex` - announced as same-family (`codex limit`) with the error quoted, because it is not an independent review.

A limit hit in the middle of a round leaves that round INCOMPLETE and prints `review: codex usage limit in round N: ....` It is recorded in `<git-common-dir>/crew/review-limit/<ticket>.json`, so the next `/crew:review` answers limited from the record without calling Codex again. Once that next round is reserved the record stops applying, and the round after it probes Codex live. The record is written after the round's verdict: when it cannot be written the round still ends INCOMPLETE (exit 3), the line says `could not record it (...)`, and the next probe calls Codex live instead.

What Is Proven, and What Is Only Configured

Run the probe. The sample below shows the probe's format with this build's values. The `hooks` feature value was measured with `codex features list`. The probe itself has only been run against a stub `codex` in the test suite (`tests/test_crew_instructions.py`).

```
python3 "$S/crew_instructions.py" codex-probe

codex: /usr/local/bin/codex (codex-cli 0.155.1)

hooks feature: enabled

project files: current

PostToolUse: configured, not proven

SessionStart: configured, not proven

SubagentStart: configured, not proven

UserPromptSubmit: configured, not proven
```

- **Established by reading the `codex 0.155.1` binary, not its docs:** the hook events (`SessionStart`, `UserPromptSubmit`, `PostToolUse`, `SubagentStart` and others), the handler keys `command`, `commandWindows`, `timeout` and `additionalContextLimit`, and a `SubagentStartHookSpecificOutputWire` output type. `codex features list` reports `hooks stable true`.

- **Not measured:** whether a hook's `additionalContext` reaches the Codex model, and whether Codex honours profiles from a project's `.codex/config.toml`. A live test run would need Codex to write its session state under `~/.codex`, and this build did not do that.
- The probe changes an event's line from "configured, not proven" to "hook invoked Nx under Codex" only when the context log records an invocation under the codex harness. Even then it claims only that the hook ran, not that the model received the context. To prove delivery, rerun the seeded-note test from [Confirming recall reaches your sessions](#) with `codex exec` in place of `claude -p`.

When Codex Is Barred

A review is independent only when it comes from a different model family than the author. If Codex wrote the diff, it cannot review that diff. The order falls through to Copilot, then to Claude's reviewer.